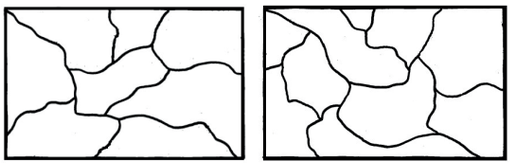
10. Colour the regions in the maps with few colours as possible but make sure that no two adjacent countries are of the same colour.



Objective Type Questions

11. In a class there are 26 boys and 15 girls. The teacher wants to select a boy or a girl to represent a quiz competition. In how many ways can the teacher make this selection?
(A) 41 (B) 26 (C) 15 (D) 390

12. How many outcomes can you get when you toss three coins once?
(A) 6 (B) 8 (C) 3 (D) 2

13. In how many ways can you answer 3 multiple choice questions, with the choices A,B,C and D?
(A) 4 (B) 3 (C) 12 (D) 64

14. How many 2 digit numbers contain the number 7 ?
(A) 10 (B) 18 (C) 19 (D) 20

7.5 Fibonacci Numbers

We have learnt in earlier classes, on how all beautiful things in nature as well as man made things are connected with Mathematics. Now, we just refresh everyone's memory and show how Math can be beautiful when seen in physical and biological things everywhere around us.

Fibonacci (real name Leonardo Bonacci) was an Italian mathematician who developed the Fibonacci Sequence. Remember the pattern of the Fibonacci sequence we already studied in standard VI it looks like this: 1, 1, 2, 3, 5, 8, 13, 21, 34... and it goes on.

Let us tabulate the Fibonacci sequence and find a rule.

Term (n)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	...
F (n)	1	1	2	3	5	8	13	21	34	55	89	144	233	377	610	...

We observe that the **3rd** term of the Fibonacci sequence is the sum of **2nd term** and the **1st term**.

That is, **F(3) = F(2) + F(1)** and so we can extend and write the rule is

F(1)	F(2)	F(3)	F(4)	F(5)	F(6)	F(7)	...
1	1	2	3	5	8	13	...

+ =

F(3) = F(2)+F(1)

F(n)= F(n–1) + F(n–2)

where **F(n)** is the **nth** term

F(n–1) is the previous term to the **nth** term

F(n–2) is the term before the **(n–1)th** term

This is how the Fibonacci Sequence is obtained. Let us learn more from the following real life examples.

8th_Maths_English Medium_Chapter 7.indd 247 03-12-2022 15:22:00